NOAH QIAO

MECHATRONICS ENGINEERING STUDENT AT THE UNIVERSITY OF WATERLOO

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TECHNICAL SKILLS

- **Mechanical**: SolidWorks, AutoCAD, TinkerCAD, SketchUp, 3D printing, Machine tools
- **Electrical & Hardware**: Altium, KiCad, LTSpice, Falstad, Soldering
- **Embedded & Controls**: Arduino, ESP32, STM32, Raspberry Pi, RTOS, PLC programming, VHDL
- **Software**: Java, Python, MATLAB, C++ (RobotC), SQL, HTML

RELEVANT EXPERIENCES

Borrum Energy Solutions
R&D Electrical - Mechatronics Engineering — Sep 2025 - Dec 2025

- Designed a wind-turbine temperature and humidity monitoring unit with SD data logging and **RTOS**-based task scheduling using **Teensy 4.1**.
- Designed a **KiCad** PCB for the temperature/humidity monitoring unit, integrating surge and ESD transient protection for harsh outdoor deployment.
- Collaborated with UofT and UBC to test and implement a microwave-based ice detection system for next-generation wind turbines.
- Automated electronic component inventory using **Python** scripts with vendor APIs, reducing manual entry by **~85%** and improving traceability.

Excelitas
Mechanical Engineering Co-op — Jan 2025 - Apr 2025

- Led a team in assembling and validating a new-generation power testing station, securing design approval and ensuring functional performance.
- Improved fixture design, resulting in a **300%** increase in testing efficiency and reduced production time.
- Created detailed sheet metal part models and manufacturing drawings for product lines using **SolidWorks**.
- Revised PCB connector layouts and schematics using **Altium** to support updated product requirements.

Sheartak Tools Ltd
Mechanical Engineering Associate — Sep 2023 - Dec 2023

- Designed and developed over 10 **SolidWorks** models for custom cutter-head profiles, enabling customer machine upgrades.
- Created detailed engineering drawings, expanding company inventory and streamlining manufacturing processes.
- Demonstrated strong mechanical expertise in assisting upgrades for customer machines, enhanced the quality of woodcutting, and improved the overall user experience.

RECENT PROJECTS

Remote Control Battle Bot — Mar 2025

- Designed and built a **remote-controlled** combat robot with an integrated spinning weapon system.
- Integrated FlySky FS2A receiver and dual-channel brushed ESC to enable wireless control.
- Conducted soldering, wiring, and troubleshooting of low-voltage DC circuits, signal connectors, and motor systems.

Gesture Controlled Robotic Hand — Aug 2024

- Designed and **3D printed** the mechanical components of a robotic hand using **SolidWorks**.
- Developed motor control systems using **Arduino Uno**, enabling precise robotic hand movements.
- Implemented gesture recognition using **Python, Mediapipe** and **OpenCV** for real-time control.

EDUCATION

University of Waterloo — Sep 2022 - Apr 2027
Candidate for Bachelor of Applied Science in Mechatronics Engineering